$-45(-\frac{4}{6}+1)$

$\frac{5}{6}$ $\frac{\frac{45}{5}}{225}$ 2

PART I: FILL IN THE BLANK SPACE ITEMS

Direction: Fill in the blank space with the most simplified answers [9%, 1.5pt each blank space]

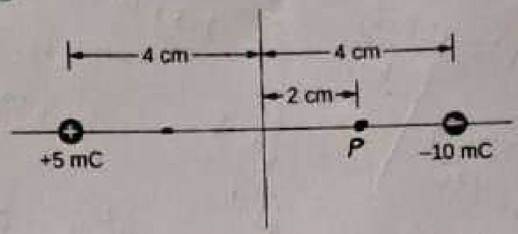
[1]. If the temperature of a copper plate with an initial area of $100cm^2$ changes from $50°c$ to $70°c$, then its change in area will be 68×10^{-7} m^2 ($take \propto = 17 \times 10^{-6} K^{-1}$)

[2]. The total electric potential at point P, due to the system of two point charges as shown in the figure below is $-225/6 \times 10^8$ V V

$V_1 = \frac{9 \times 10^9 \times 5 \times 10^{-3}}{4 \times 10^{-2}}$

$45/6 \times 10^{-8}$ -45 $-\frac{9}{2}$

$V_2 = \frac{9 \times 10^9 \times -10^{-3}}{2 \times 10^{-2}}$



[3]. Two resistors connected in series have an equivalent resistance of 690 Ω. When they are connected in Parallel, their equivalent resistance is 150Ω. The resistance of the two resistors are 469.5Ω and 220.4Ω

[4]. If the magnitude of the electric force between two protons is 2.3×10^{-26} N. The distance r between the two protons will be ~~[illegible]~~ m (Take charge of a proton as $1.6 \times 10^{-19}c$, and $K = 9 \times 10^9 \, Nm^2/C^2$)
↳ 0.10m

[5]. The small piston of a hydraulic lift has an area of $0.05 \, m^2$. A car weighing 1200N sits on a rack mounted on the large piston. The large piston has an area of $0.5m^2$. The force that must be applied to the small piston to support the car will be 120N N.

[6]. The net work done by the gas undergoing the cyclic process illustrated in Figure below is 4×10^5 J.

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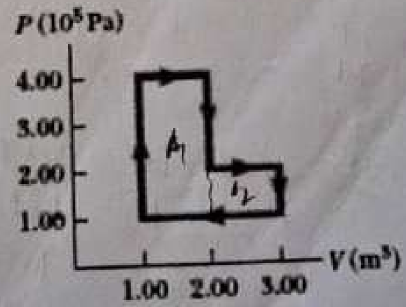
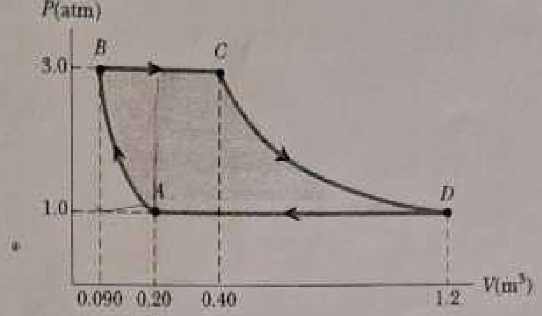


150
101.3
48.7

A sample of an ideal gas goes through the process shown in the Fig below. From A to B, the process is adiabatic; from B to C, it is isobaric with 100 kJ of energy entering the system by heat. From C to D, the process is isothermal; from D to A, it is isobaric with 150 kJ of leaving the system by heat. Determine the difference/change in internal energies for the three processes **(5pts.)**

(a) ΔU_{CD}.
(b) ΔU_{AD}
(c) ΔU_{AB}



~~(a) ΔU_{CD}~~

~~(b) $\Delta U_{AD} = -150\times10^3 + (101\times10^3$~~ $(1))$

5/5

~~b) ΔU_{AD}:~~

a) $\Delta U_{CD} = 0$, since $\Delta T = 0$ for isothermal process

b) $\Delta U_{AD} = (-150\times10^3 + (101.3\times10^3 (1))$

$= (48.7\times10^3)]$

c) $\Delta U_{AB} = W_{AB}$ $\quad P_A = \frac{nRT_A}{V_A} \Rightarrow T_A = \frac{P_A V_A}{nR}$

$= \frac{3}{2} nR\Delta T$ $\quad T_B = \frac{P_B V_B}{nR}$

$\frac{3}{2} nR(T_B - T_A)$

$\frac{3}{2} \frac{nR}{nR}\left(\frac{P_B V_B}{1} - P_B V_B\right)$

$= \frac{3}{2} (303.9\times10^3 (0.09) - (101.3\times10^3 \times (0.2)))$

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$A_T = 1208 \times 10^{-6}$ $P = \frac{F}{A} = \frac{mg}{A}$ $= \frac{66 \times 10}{1208 \times 10^{-6}}$ 0.546×10^{6} 5.46×10^{5}

D

[5]. A 66 kg man lies on his back on a bed of nails with 1208 of the nails in contact with his body. What is the total pressure exerted on the man's body if the tip of each nail has an area of 10^{-6} m^2? (*Take* g=10 m/s^2)

A. 2.21×10^5 Pa B. 3.09×10^5 Pa C. 1.65×10^5 Pa D. 5.45×10^5 Pa E. 4.11×10^4 Pa

A

[6]. A solid has a volume of 8 cm^3. When weighed in air on a spring scale calibrated in grams, the scale indicates 20g. What does the scale indicates if the object is weighed while it is immersed in a liquid of density 2g/cm^3?

W_{air}: 20g
$F_b = \rho g V$
$= 2 \times 10 \times 8$
$= 160$

A. 4g B. 10g C. 12g D. 16g E. 2g

D

[7]. For the given three vectors, $\vec{A} = 3i - 2j + 4k$, $\vec{B} = 2i + 2j - 3k$ and $\vec{C} = -i - 4j + k$, then the magnitude of the resultant vector is:

A. $3\sqrt{3}$ B. 2 C. $4\sqrt{2}$ D. 6

B

[8]. A stone is thrown with an initial velocity of 20 $\frac{m}{s}$ at an angle of 30^0, the maximum height the stone will rise is (Take sine $30^0 = 0.5$, g= 10 $\frac{m}{s^2}$):

A. 15m B. 5 m C. 10.2 m D. 3.9 m E. None

A

[9]. The current in a circuit loop that has a resistance R_1 is 2.0 A. The current is reduced to 1.6 A when an additional resistor of resistance $R = 3\Omega$ is in series with R_1. What is the value of R_1?

$\frac{V}{R_1} = 2$ / $\frac{V}{R_1+3} = 1.6$
$2R_1 = 1.6R_1 + 4.8$
$0.4R_1 = 4.8$
$R_1 = 12$

A. 12Ω B. 4.8Ω C. 2.4Ω D. $3/4$ Ω

B C

[10]. Suppose a point charge $q_1 = -9\mu C$ is at x=0 while $q_2 = 4\mu C$ is at x=1m. At what point would the total electrostatic force on a positive charge $q_3 = 1\mu C$ be zero with respect to the origin?

A. x=2m B. x=3m C. x=4m D. x=1.5m E. none of the above

$F_b = 160$ $F_b = W_{air} - W_{app}$
$W_{app} = F_b - W_{air} = 160 - 200$
$\frac{40}{10} = 4g$

$4i - 4j + 2k$ $\sqrt{16+16+4}$ $\sqrt{36} = 6$

$h_{max} = \frac{u^2 \sin^2\theta}{2g}$ $v^2 = u^2 + 2as$
$\frac{20^2 (\sin 30)^2}{2 \times 10}$
$= \frac{400 \times 0.25}{20}$
$= 5$

$h_{max} = \frac{u_y^2}{2g}$

$R_T = R_1 + R_2 = R_1 + 3$ $I = \frac{V}{R}$
$R_1 \cdot 2 = V$
$V = 2R_1$
$I_2 = \frac{2R_1}{R_1+3}$

$\frac{V}{R_1} = 2$ and $\frac{V}{R_1+3} = 1.6$
$V = 2R_1$ and $V = (R_1+3)1.6$
$2R_1 = (R_1+3)1.6 \Rightarrow R_1 = 12$

$\frac{4}{R_1} = 2$ $\frac{4}{R_1+3} = 1.6$
$2R_1 = (R_1+3)1.6$
$0.4R_1 = 3$

$x = 1+2$
$\frac{9}{(x+1)^2} = \frac{4}{x^2}$
$F = k\frac{q_1 q_3}{r^2}$ $= \frac{k q_1 q_3}{(x+1)^2} = \frac{k q_2 q_3}{x^2}$
$4x^2 = 4x^2 + 8x + 4$
$5x^2 - 8x - 4 = 0$

$2(4I_1+3I_2=12) = 8I_1+6I_2=24$

$-3(-6I_1+2I_2=-5)$ $\quad 18I_1-6I_2=15$

$26I_1=39$

$I_1=\frac{39}{26}$

[5]. Find the values of the currents I_1, I_2, I_3, I_4 and I_5 in the circuit shown in the figure below. **Assume the direction of the loop is Clockwise: (6Pts.)**

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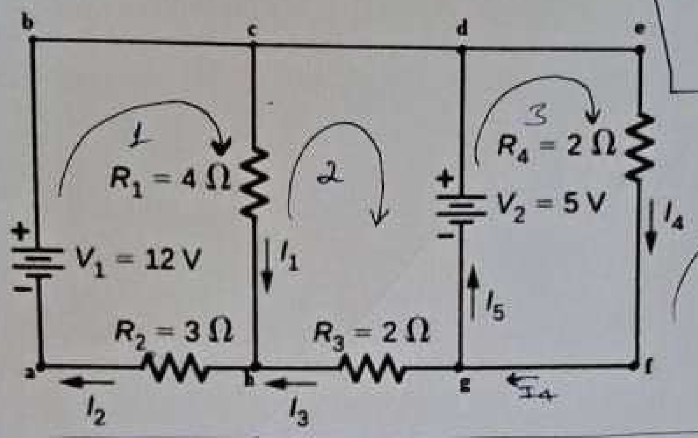

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From Kirchhoff's Junction rule:

1) $I_2 = I_1 + I_3 \Rightarrow I_3 = I_2 - I_1$

2) $I_4 = I_5 + I_3$

apply Kirchhoff's loope rule

loope 1

$12V = 3I_2 + 4I_1$

$\Rightarrow 4I_1 + 3I_2 = 12 \cdots (1)$

loope 2

$-5V = 2I_3 - 4I_1$

$\Rightarrow 2I_3 - 4I_1 = -5 \cdots (2)$

$\Rightarrow 2I_3 = -5 + 4I_1$

$I_3 = \frac{-5+4I_1}{2}$

$\frac{-5+4(2)}{2} \quad \frac{3}{2}$

loope 3

$5V = 2I_4$

$\Rightarrow 2I_4 = 5 \cdots (3)$

$\boxed{I_4 = \frac{5}{2} = 2.5A}$

Solve those eqns

$\Rightarrow I_3 = I_2 - I_1$

$\Rightarrow 2(I_2 - I_1) - 4I_1 = -5 \cdots$ eqn (2)

$\Rightarrow 2I_2 - 2I_1 - 4I_1 = -5$

$\Rightarrow 2I_2 - 6I_1 = -5 \cdots (4) \leftrightarrow$ with eqn (1)

$$\Rightarrow \left\{\begin{pmatrix} 4I_1 + 3I_2 = 12 \\ -6I_1 + 2I_2 = -5 \end{pmatrix}\begin{matrix} 2 \\ -3 \end{matrix} \Rightarrow \begin{cases} 8I_1 + 6I_2 = 24 \\ 18I_1 - 6I_2 = 15 \end{cases}\right.$$

$26I_1 = 39$

$\Rightarrow \boxed{I_1 = \frac{39}{26} = 1.5A}$

[10]

$\Rightarrow 4I_1 + 3I_2 = 12 \Rightarrow 3I_2 = 12 - 6 = 6$

$\Rightarrow \boxed{I_2 = \frac{6}{3} = 2A}$

$\Rightarrow I_3 = I_2 - I_1 = 2 - 1.5 = 0.5$

$\Rightarrow I_4 = I_5 + I_3 \Rightarrow I_5 = I_4 - I_3$

$= 2.5 - 0.5$

$= 2A$

Finally

$I_1 = 1.5A$

$I_2 = 2A$

$I_3 = 0.5A$

$I_4 = 2.5A$

$I_5 = 2A$

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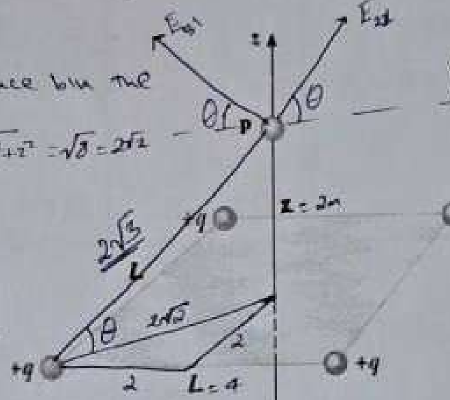
[4]. Four identical charges, each having a charge of $q=+2\mu C$, are fixed at the corners of a square of side $L=4m$ as shown in figure below.

a. Find the electric field at a point P that lies at a distance $Z = 2m$ along the line perpendicular to the plane of the square and passing through the center of the square.**(3Pts.)**

b. Find the net electric force on a fifth point charge of $Q=-4\mu C$, at a point P that lies at $Z=2m$ above the center of the square due to the other four charges.**(3Pts.)**



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first let's get the distance b/w the two charges.

$\frac{L}{2} = 2 \Rightarrow \sqrt{2^2+2^2} = \sqrt{8} = 2\sqrt{2}$

$\Rightarrow d \rightarrow d = \sqrt{8+4} = \sqrt{12} = \underline{2\sqrt{3}}$

2

direction?

ⓐ

$E = \sum \frac{kq}{d^2} = \left(\frac{9\times10^9 \times 2\times10^{-6}}{12}\right)4 \sin\theta \Rightarrow (1.5\times10^3)4 \cdot \frac{2}{2\sqrt{3}}$

$\Rightarrow \boxed{\left(\frac{kq}{d^2}\right)4 \cdot \sin\theta}$ - only the Y-component (Z-component)

2/3

$\Rightarrow 6\times10^3 \cdot \frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$

$\Rightarrow 6 \cdot \frac{\sqrt{3}}{3} \times 10^3$

$\boxed{\Rightarrow 3.46 \times 10^3}$

ⓑ $F = E \cdot q = (3.46\times10^3 \times 4\times10^{-6})$

$= \underline{13.8 \times 10^{-3}}$

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PART II: SHORT ANSWER ITEMS

Direction: Give short answers for the following questions only on the space provided [6%, 1pt each]

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[1]. Define the following terms:-

i) Specific heat capacity

* specific heat capacity is the heat energy required to raise ~~1~~ $\underline{1\,kg}$ of a body by $\underline{1^{o}C}$.

ii) Latent heat of fusion

* Latent heat of fusion is the heat absorbed or realesed when a matter change state from solid to liquid at a $\underline{\text{Constant Temprature}}$.

$\boxed{Q = mL_{fusion}}$

[2]. State and explain the following laws:

i) The Zeroth law of thermodynamics

"If an object A is in termal equilibrium with object B, and object B is in termal equilibrium with object C, then object A and object C are in termal equilibrium."

A C B

$T_A = T_B$, $T_B = T_C \Rightarrow \underline{\underline{T_A = T_C}}$

ii) Kirchhoff's Junction rule

→ It states that "the amount of current entering a junction is equal to the current leaving the junction" $\underline{\text{"Conservation of Charge"}}$

$I_1 = I_2 + I_3$

iii) Faraday's Law of electromagnetic induction

"the strength of emf induced in a circuite is directly proportional to the rate of change of magnetic flux."

$\Rightarrow \boxed{Emf = N \frac{\Delta \phi}{\Delta t}}$

[3]. List two heat transfer mechanisms and explain their differences

There are 3 heat transfer mechanisms, these are: Conduction, Convention, and Radiation

⇒ 1. <u>Conduction</u>:- is the transfer of heat energy by physical (direct) Contact.

2. <u>Convension</u>: is the transfer of heat energy through a mediator or a medium of transmition (i.e air molecules, water molecules)

3. <u>Radiation</u>:- it is a transfer of heat energy that doesnot require any medium of transmition (i.e can travel through vacuum) eg:- sun radiation.

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[2]. Calculate the total amount of heat energy required to convert a **1g** cube of ice at **- 30°C** to steam at **120 °C**. (Use c_{ice} =2090 J/Kg°C, L_{fusion}=3.33x10^5 J/Kg, c_{water} =4.19 x10^3 J/Kg°C, L_{vapor}=2.26x10^6 J/Kg, c_{steam} =2.01 x10^3 J/Kg°C) **(4Pts.)**

4/4

$\Rightarrow Q_{ice} = m C_{ice} \Delta T \Rightarrow 1\times10^{-3} \times 2090 \times (0° - (-30°C)) \Rightarrow 62700 \times 10^{-3} J$
$= 62.7 J$

$Q_{fusion} = m L_{fusion} = 1\times10^{-3} kg \; 3.33\times10^5 J/kg = 3.33\times10^2 J = 333 J$

$Q_{water} = m C_{water} \Delta T = 1\times10^{-3} kg \; 4.19\times10^3 J/kg°C \cdot (100° - 0°)°C = 419 J$

$Q_{vapor} = m L_{vapor} = 1\times10^{-3} kg \times 2.26\times10^6 J/kg = 2.26\times10^3 J$
$= 2260 J$

$Q_{steam} = m C_{steam} \Delta T = 1\times10^{-3} kg \times 2.01\times10^3 J/kg°C \times (120 - 100)$
$= 40.2 J$

$\Delta Q = Q_{ice} + Q_{fusion} + Q_{water} + Q_{vapor} + Q_{steam}$

$= 62.7J + 333J + 419J + 2260J + 40.2J$

$= 3114.9 J$

1. $F = q(v \times B) \rightarrow$

	i	j	k
v	3×10^5	0	0
B	0	0	0.8

$0 - (0.8(3\times10^5) - 0)j + 0$

$= -2.4\times10^5 j$

$= -3.84\times10^{-14} \approx -4\times10^{-14}$

PART III: MULTIPLE CHOICE ITEMS

Direction: choose the correct answer and write the letter of your choice only on the space provided. [10%, 1 point each]

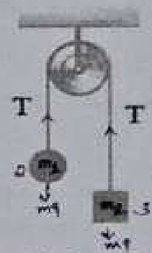
E [1]. A proton of charge ($q = 1.6\times10^{-19}$C) is moving at 3×10^5m/s in the positive x direction. A magnetic field of 0.8T is in the positive z direction. The magnetic force on the proton is:

A. 0 B. 4×10^{-14} N, in the positive z direction C. 4×10^{-14} N, in the negative z direction

D. 4×10^{-14} N, in the positive y direction E. 4×10^{-14} N, in the negative y direction

A [2]. Two objects of masses m_1=2kg and m_2=3kg are connected by a light, inextensible cord and hung over a frictionless pulley as shown. Assuming that the cord and pulley are massless, the tension **T** in the cord is: (Take g= 10 $\frac{m}{s^2}$)

A. 24*N* B. 8.75*N* C. 42*N* D. 17*N* E. None



$T - m_1 g = m_1 a$

$m_2 g - T = m_2 a$

$m_2 g - m_1 g = a(m_1 + m_2)$

$a = \frac{m_2 g - m_1 g}{m_1 + m_2} = \frac{10}{5} = 2\,m/s^2$

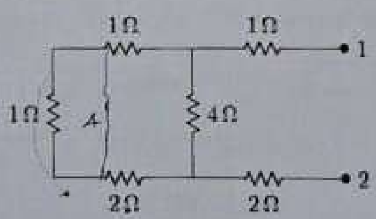
$T - m_1 g = m_1 a$

$T = 24\,N$

$\frac{4\times4}{4+4} = \frac{16}{8} = 2$

C [3]. The equivalent resistance between points 1 and 2 of the circuit shown is:

A. 3 Ω B. 4 Ω C. 5 Ω D. 6 Ω E. 7 Ω



B. [4]. A 1.0-mol sample of an ideal gas is kept at a constant temperature of 0 °C during an expansion from 3 L to 10 L. How much work is done by the gas during the expansion? (Take $R = \frac{8.31 J}{mol.K}$)

A. 2.7×10^3 J B. -2.7×10^3 J C. 1.6×10^3 J D. -1.6×10^3 J

$W = nRT\ln\left(\frac{V_i}{V_f}\right)$

$= 1\times8.31\times273\times\ln\left(\frac{3}{10}\right)$

$= 2268.63\ln\left(\frac{3}{10}\right) = 2268.63\times(-1.20397)$

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PART IV: WORKOUT ITEMS

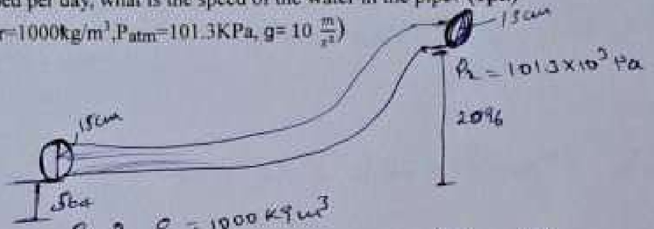
Direction: Solve the following problems by showing all the necessary steps briefly only on the space provided [25%]

[1]. Through a pipe 15.0 cm in diameter, water is pumped from the a River up to a Village, located on the rim of the canyon. The river is at an elevation of 564 m, and the village is at an elevation of 2096 m.

(a) What is the minimum pressure at which the water must be pumped if it is to arrive at the village? **(3pts.)**

(b) If 4500 m^3 are pumped per day, what is the speed of the water in the pipe? **(1pt.)**

(Use density of water=1000kg/m^3, P_{atm}=101.3KPa, g= 10 $\frac{m}{s^2}$)



$P_1 = ?$ $\rho = 1000$ kg/m^3

101300 + 1532

Continuity eqn

$A_1V_1 = A_2V_2$ since the diameter is constant i.e. $A_1 = A_2 \Rightarrow V_1 = V_2$ ⊕ $Q_1 = Q_2$

$\Rightarrow P + \frac{1}{2}\rho V^2 + \rho g h = \text{Constant} \Rightarrow P_1 + \frac{1}{2}\rho V_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho V_1^2 + \rho g h_2$

$\Rightarrow P_1 = P_2 + \rho g h_2 - \rho g h_1$

$= 101.3\times10^3 + 1000\times10\,(h_2 - h_1)$

$2096 - 564 = 1532$

$= 101.3\times10^3 + 10000\,(1532)$

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(a) $\rightarrow P_1 = \underline{15.4 \text{ MPa}}$

$r = d/2$

(b) the flow rate = 4500 m^3/day

1 day = 24h, 1h = 3600 sec, 24h = ? ⇒ 86400 sec

$\boxed{Q = AV} \rightarrow \pi r^2 . V = 4500\ m^3/day$

$\pi\left(\frac{15\times10^{-2} m}{2}\right)^2 . V = \frac{4500 m^3}{86400 sec} = 0.0521\ m^3/sec$

$\frac{0.052}{\pi\times 225\times10^{-4}}$

$\frac{0.052}{706.5}\times10^4$

$V = \frac{0.0521\times4}{225\pi\times10^{-4}}$

$\Rightarrow V = \underline{3\ m/s}$